

Automated EDA + Data Quality Detection with LLM-Assisted Reporting: Literature & Tool Review

Section 1 — Academic Papers

1.1 Foundational Works

P1. Profiling Relational Data: A Survey

- **Authors:** Ziawasch Abedjan, Lukasz Golab, Felix Naumann
- **Year/Venue:** 2015, *The VLDB Journal* 24(4): 557–581
- **DOI:** <https://doi.org/10.1007/s00778-015-0389-y>
- **Problem statement:** How can metadata (statistics, dependencies, uniqueness, data types) be systematically extracted from relational datasets without requiring user-specified queries?
- **Approach:** Defines the full taxonomy of data profiling tasks—single-column analysis (distributions, data types, cardinality, null counts), two-column analysis (correlations, functional dependencies), and multi-column analysis (inclusion dependencies, unique column combinations)—and surveys algorithms for each. Proposes a unified profiling language (DPQL) and cost models for profiling at scale.
- **Relevance:** Canonical reference for Layer-1 design; the taxonomy of single/multi-column profiling directly maps to what the project's deterministic profiling engine must compute. HIGH
- **Tag:** [verified][¹][2]

P2. Data Quality Toolkit: Automatic Assessment of Data Quality and Remediation for Machine Learning Datasets

- **Authors:** Nitin Gupta, Hima Patel, Shazia Afzal et al. (IBM Research)
- **Year/Venue:** 2021, arXiv:2108.05935 [cs.LG]
- **Link:** <https://arxiv.org/abs/2108.05935>
- **Problem statement:** Standard data cleaning tools do not detect ML-specific data quality issues like class overlap, feature leakage, or noisy labels.
- **Approach:** Introduces the IBM Data Quality for AI Toolkit as a library of DQ metrics (completeness, consistency, accuracy, uniqueness, label purity, class overlap) plus remediation functions. Each metric is assessed independently, and results are output via structured APIs. Provides tutorials via IBM Learning Path and public API Hub.
- **Relevance:** Closest existing work to the project's Layer 2 + Layer 3 concept: structured, ML-readiness-oriented DQ metrics with an API-accessible result schema. Also defines the "DQ for ML" problem space the project targets. HIGH
- **Tag:** [verified][¹][3]

P3. ADBench: Anomaly Detection Benchmark

- **Authors:** Songqiao Han, Xiyang Hu, Hailiang Huang, Minqi Jiang, Yue Zhao
- **Year/Venue:** 2022, *NeurIPS 2022 Datasets and Benchmarks Track*
- **Link:** <https://arxiv.org/abs/2206.09426>
- **Problem statement:** How do 30 existing tabular anomaly detection algorithms compare across levels of supervision, anomaly type, and data corruption?
- **Approach:** Runs 98,436 experiments across 57 tabular benchmark datasets using 30 algorithms spanning unsupervised, semi-supervised, and supervised paradigms; open-sources both datasets and a plug-and-play testbed. Reveals that supervision level dominates performance differences more than algorithm choice for most tabular tasks.
- **Relevance:** The de-facto benchmark for selecting and justifying Layer-2 outlier detection algorithm choices (IsolationForest, LOF, ECOD, COPOD), directly cited by PyOD documentation. HIGH
- **Tag:** [verified][⁴][5][⁶]

P4. ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions

- **Authors:** Zheng Li, Yue Zhao, Xiyang Hu, Nicola Botta, Cezar Ionescu, George H. Chen
- **Year/Venue:** 2022, *IEEE Transactions on Knowledge and Data Engineering (TKDE)*
- **Link:** <https://arxiv.org/abs/2201.00382>

- **Problem statement:** Existing unsupervised outlier detection methods suffer from high computational cost, hyperparameter tuning complexity, and limited interpretability for high-dimensional tabular data.
- **Approach:** ECOD estimates the underlying data distribution non-parametrically using empirical cumulative distribution functions per dimension, then aggregates tail-probability estimates to produce an outlier score. It is parameter-free and provides feature-level interpretability (which dimension drove the score). Outperforms 11 baselines across 30 datasets in accuracy, efficiency, and scalability. Implemented in PyOD.
- **Relevance:** Directly implements one of the project's named Layer-2 algorithms; its parameter-free design and interpretable per-dimension score are ideal for populating Layer-3 statistical_basis fields. HIGH
- **Tag:** [verified]^{[7][8]}

P5. PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection

- **Authors:** Sihan Chen, Zhuangzhuang Qian, Wingchun Siu et al. (USC)
- **Year/Venue:** 2024, arXiv:2412.12154 [cs.LG]
- **Link:** <https://arxiv.org/abs/2412.12154>
- **Problem statement:** The original PyOD library lacked modern deep learning models, had fragmented PyTorch/TF implementations, and provided no automated model selection, creating barriers for non-expert users.
- **Approach:** PyOD 2 integrates 12 state-of-the-art deep learning OD models into a unified PyTorch framework and adds an LLM-based pipeline (ADEngine) for automated model selection from the 45-algorithm library, guided by ADBench benchmark results.
- **Relevance:** The definitive paper for PyOD (primary Layer-2 library); the LLM-for-model-selection pattern (LLM picks algorithm, does not compute statistics) is a close analogue to the project's Layer-4 "interpreter not analyst" principle. HIGH
- **Tag:** [verified]^{[9][10][^11]}

P6. Data-Centric Artificial Intelligence: A Survey

- **Authors:** Daochen Zha, Zaid Pervaiz Bhat, Kwei-Herng Lai, Fan Yang, Zhimeng Jiang, Shaochen Zhong, Xia Hu
- **Year/Venue:** 2023, arXiv:2303.10158 [cs.AI]
- **Link:** <https://arxiv.org/abs/2303.10158>
- **Problem statement:** How does one systematically understand and organize the rapidly growing body of methods and tools for improving data quality and quantity across the full ML data lifecycle?
- **Approach:** Proposes a taxonomy of data-centric AI tasks under three goals—training data development, inference data development, and data maintenance—covering data collection, labeling, preparation, augmentation, quality evaluation, drift monitoring, and more. Surveys representative methods in each category and benchmarks.
- **Relevance:** Positions the project within the emerging data-centric AI movement; the "data maintenance" pillar directly maps to Layers 1–3. Establishes the intellectual context for why deterministic DQ is now a first-class ML concern. HIGH
- **Tag:** [verified]^{[12][13]}

P7. Data Quality Awareness: A Journey from Traditional Data Management to Data Science Systems

- **Authors:** Sijie Dong, Soror Sahri, Themis Palpanas
- **Year/Venue:** 2024, arXiv:2411.03007 [cs.DB]
- **Link:** <https://arxiv.org/abs/2411.03007>
- **Problem statement:** How has data quality awareness evolved from classical database management to modern ML/big data systems, and what new DQ challenges have emerged?
- **Approach:** Reviews DQ techniques across traditional DBMS, big data systems, and ML pipelines using a cause-effect model linking big data quality challenges to emerging ML DQ issues. Covers DQ dimensions, anomaly detection, schema management, and data profiling in each era. Claimed to be the first cross-era DQ survey.
- **Relevance:** Provides historical grounding for Layer-1 and Layer-3 design and clarifies which classical DQ techniques (constraint checking, functional dependency discovery) remain relevant for ML-readiness assessment. HIGH
- **Tag:** [verified]^{[14][15]}

P8. A Survey on Data Quality Dimensions and Tools for Machine Learning

- **Authors:** Yuhan Zhou, Fengjiao Tu, Kewei Sha, Junhua Ding, Haihua Chen
- **Year/Venue:** 2024, arXiv:2406.19614 [cs.DB]
- **Link:** <https://arxiv.org/abs/2406.19614>

- **Problem statement:** Which DQ dimensions and metrics are most relevant for ML, and how well do the 17 major DQ tools cover them?
- **Approach:** Defines four DQ dimensions (intrinsic, contextual, representational, accessibility) with associated ML-oriented metrics, then systematically audits 17 tools (including GE, DeeQu, whylogs, TFDV, deepchecks) against these dimensions. Finds that most tools emphasize intrinsic/representational dimensions with weak contextual (ML-readiness) coverage.
- **Relevance:** Directly informs Layer-3 ontology design: the dimension taxonomy is a candidate schema for categorizing DQ findings. The coverage gap for ML-contextual dimensions is the project's primary differentiator. **HIGH**
- **Tag:** [verified]^{[16][17][^18]}

P9. A Systematic Review of Tools for AI-Augmented Data Quality Management in Data Warehouses

- **Authors:** Heidi Carolina Tamm (Martinsaari), Anastasija Nikiforova
- **Year/Venue:** 2024, *Business Informatics Research Conference (BIR 2025)*; arXiv:2406.10940 [cs.DB]
- **Link:** <https://arxiv.org/abs/2406.10940>
- **Problem statement:** To what extent do existing DQ tools support AI-augmented automation of rule detection and anomaly identification in data warehouses?
- **Approach:** Systematic review of 151 DQ tools using multi-phase screening (functionality, trialability, GDPR, architecture); only 10 tools met the full criteria for AI-augmented DQM. Finds that most tools focus on data cleansing for AI rather than using AI to improve DQ management itself; explainability features are scarce.
- **Relevance:** Empirical confirmation of the Layer-4 gap: almost no tools currently implement AI-as-interpreter over deterministic findings. Provides tool comparison matrix useful for positioning. **HIGH**
- **Tag:** [verified]^{[19][20][^21]}

P10. AD-LLM: Benchmarking Large Language Models for Anomaly Detection

- **Authors:** Tiankai Yang, Yi Nian, Shawn Liet al. (USC/Northwestern/Adobe)
- **Year/Venue:** 2024, arXiv:2412.11142; accepted *Findings of ACL 2025*
- **Link:** <https://arxiv.org/abs/2412.11142>
- **Problem statement:** How effectively can LLMs assist with anomaly detection tasks—zero-shot detection, data augmentation, and model selection—compared to traditional approaches?
- **Approach:** Introduces the AD-LLM benchmark evaluating GPT-4 and Llama 3.1 on NLP AD tasks across three modalities. Finds LLMs are effective at zero-shot AD and data augmentation but that explaining model selection for specific datasets remains challenging. Proposes six future research directions.
- **Relevance:** Illuminates the boundary between what LLMs can usefully contribute (model selection guidance, natural language reasoning about anomalies) versus what they cannot reliably do autonomously. Informs the rationale for the Layer-4 "interpreter not detector" architecture. **MEDIUM**
- **Tag:** [verified]^{[22][23][^24]}

P11. LLMs for Explainable AI: A Comprehensive Survey

- **Authors:** Ahsan Bilal, David Ebert, Beiyu Lin
- **Year/Venue:** 2025, arXiv:2504.00125 [cs.AI]
- **Link:** <https://arxiv.org/abs/2504.00125>
- **Problem statement:** How can LLMs transform complex ML model outputs (feature importances, SHAP values, counterfactuals) into accessible natural-language explanations for non-expert users?
- **Approach:** Surveys the landscape of LLM-for-XAI approaches covering template-to-LLM evolution, evaluation metrics for explanations (faithfulness, coherence, user satisfaction), and real-world applications across healthcare, finance, and operations. Reviews hallucination risks in explanation generation.
- **Relevance:** The XAI-to-narrative pattern is the closest academic analogue to the project's Layer-4 pattern (deterministic signal → LLM narrative). Specifically covers faithfulness metrics that should constrain Layer-4 outputs. **MEDIUM**
- **Tag:** [verified]^[^25]

P12. A Review of Faithfulness Metrics for Hallucination Assessment in LLMs

- **Authors:** Ben Malin, Tatiana Kalganova, Nikoloas Boulgouris
- **Year/Venue:** 2024 (published 2025), arXiv:2501.00269 [cs.CL]
- **Link:** <https://arxiv.org/abs/2501.00269>
- **Problem statement:** How should faithfulness (factual consistency with a grounding source) be evaluated in LLM-generated text, and which mitigation strategies are most effective?

- **Approach:** Reviews faithfulness evaluation metrics across summarization, QA, and MT tasks; identifies RAG-based and structured prompting frameworks as the most effective mitigation strategies for hallucination. Discusses why LLM-as-evaluator correlates best with human judgment over automated metrics.
- **Relevance:** Directly applicable to Layer-4 design: the review identifies that **structured input with explicit factual anchors (Layer-3 findings)** reduces hallucination, which is the architectural bet of the project. **HIGH**
- **Tag:** [verified][^26]

P13. Automated Data Preparation for Machine Learning: A Survey

- **Authors:** Sasa Mladenovic, Marius Lindauer, Carola Doerr
- **Year/Venue:** 2025, *AutoML Workshop*; available on HAL / OpenReview
- **Link:** <https://openreview.net/forum?id=Euti6LHIOs>
- **Problem statement:** Data preparation—the most time-consuming stage of the ML pipeline—is largely overlooked by AutoML systems, which assume data is already clean and formatted.
- **Approach:** Claimed to be the first survey focusing explicitly on automated data preparation as a component of AutoML. Covers transformation sequencing, imputation automation, type inference, encoding, and integration with end-to-end AutoML frameworks. Identifies the gap between data-centric AI and AutoML.
- **Relevance:** Positions the project in the AutoML/AutoDS landscape; demonstrates that Layers 1–2 solve a currently under-automated part of the ML pipeline. **MEDIUM**
- **Tag:** [verified][²⁷][28]

P14. LLM/Agent-as-Data-Analyst: A Survey (*LOW priority — landscape contrast*)

- **Authors:** Multiple; arXiv:2509.23988 [cs.DB] (September 2025)
- **Link:** <https://arxiv.org/abs/2509.23988>
- **Problem statement:** How have LLMs and agent frameworks transformed data analysis tasks across structured, semi-structured, unstructured, and heterogeneous data?
- **Approach:** Surveys NL2SQL, ModelQA, table QA, chart understanding, and agentic pipeline orchestration; distills four design goals for intelligent data analysis agents.
- **Relevance:** Canonical reference for the landscape the project explicitly distinguishes itself from (LLM-as-analyst vs. LLM-as-interpreter). Useful for the "what we are NOT" positioning section. **LOW**
- **Tag:** [verified][^29]

Note on Abedjan VLDB 2015 DOI: The paper is consistently cited as DOI 10.1007/s00778-015-0389-y across multiple independent sources. No conflicting arXiv ID was found; the paper predates arXiv-first norms for database venues and was published directly in the VLDB Journal.^[30][31][²][1]

Note on DAMA-DMBOK / ISO 8000 / ISO 25012: These are standards bodies, not arXiv papers, and are therefore not listed in this section. DAMA-DMBOK's six DQ dimensions (Completeness, Consistency, Integrity, Timeliness, Validity, Uniqueness) are referenced across the tools in Section 2 and discussed in Section 4. The DAMA NL whitepaper on Dimensions of Data Quality (DDQ, v1.2, Sept 2020) is verified as a freely accessible PDF at <https://www.dama-nl.org/wp-content/uploads/2020/09/DDQ-Dimensions-of-Data-Quality-Research-Paper-version-1.2-d.d.-3-Sept-2020.pdf>.^[32][33]

Section 2 — GitHub Repositories

PRIMARY — Classical EDA / Profiling Libraries

Name	URL	Stars (May 2026)	One-line description	Activity	Input formats	Output formats	Strengths vs project
ydata- profiling	https://github.com/ydataai/ydata-profilin g	~13,000	One-line EDA for Pandas/Spark DataFrames with HTML + JSON export	Active (YData maintains actively)	CSV, Parquet, Pandas DF, Spark DF	HTML, JSON, Jupyter widget	Rich JSON export (to_json()) containing per- column distribution stats, correlations, missing values, duplicates — excellent Layer-1 baseline; Spark support for scale

Name	URL	Stars (May 2026)	One-line description	Activity	Input formats	Output formats	Strengths vs project
Sweetviz	https://github.com/fbdesignpro/sweetviz	~3,000	Beautiful comparative EDA reports (train vs. test) in two lines	Moderately active	CSV, Pandas DF	HTML (self-contained), Python object	Train/test comparison baked in — useful for detecting dataset shift at EDA time
DataPrep	https://github.com/sfu-db/dataprep	~2,000	Low-code EDA, data cleaning, and connector library from SFU	Active	CSV, Pandas, Dask DF	HTML interactive report, Python objects	Dask-backed, ~10× faster than ydata-profiling on large DFs; DataPrep.clean for standardization
AutoViz	https://github.com/AutoViML/AutoViz	~1,500	Automatic chart generation for any dataset	Moderately active	CSV, Pandas DF	Interactive charts (Bokeh/Matplotlib), HTML	Rapid visual EDA
klib	https://github.com/akanzi/klib	~1,000	Lightweight data cleaning and visualization helpers for Pandas	Moderately active	Pandas DF	Matplotlib figures, cleaned DF	Simple, composable cleaning functions
D-Tale	https://github.com/man-group/DTale	~4,500	Flask/React interactive UI for exploring Pandas DataFrames	Active	Pandas DF, Xarray, Dask	Interactive HTML UI, Python/JSON API	Interactive column-level analysis; can export statistics as JSON via API
Capital One DataProfiler	https://github.com/capitalone/DataProfiler	~1,400	Profile + PII/sensitive-data detection with a built-in deep learning labeler	Active	CSV, Parquet, JSON, AVRO, Text, Pandas DF	JSON report (<code>profile.report()</code>), HTML	Unique: built-in PII entity detection via NER model; JSON report schema includes <code>global_stats</code> + per-column <code>data_stats</code> dictionaries — directly usable as Layer-1 output

PRIMARY — Deterministic DQ Frameworks

Name	URL	Stars (May 2026)	One-line description	Activity	Input formats	Output formats
Great Expectations	https://github.com/great-expectations/great_expectations	~11,400	"Unit tests for data" — define Expectations, validate, get JSON results	Active	CSV, Parquet, SQL DB, Spark, Pandas	HTML (Data Docs), JSON (ExpectationValidationResult), Python dict
Deequ	https://github.com/awslabs/deequ	~3,300	Amazon's Spark-native "unit tests for data" with automated constraint suggestion	Active (AWS-maintained)	Apache Spark DataFrames (via pydeequ for Python)	Spark DataFrames (constraint check results), Scala case classes, JSON via conversion
pandera	https://github.com/unionai-oss/pandera	~3,800	Pythonic schema validation for Pandas/Polars/Spark DataFrames using type annotations	Active	Pandas, Polars, Modin, Dask, Spark DF	Python SchemaError exceptions, report objects
whylogs	https://github.com/whylabs/whylogs	~2,700	Statistical profiling library producing compact, mergeable profiles for ML data	Active (WhyLabs-maintained)	Pandas DF, PySpark, images, text	Binary protobuf profile (mergeable), summary, Parquet export

Name	URL	Stars (May 2026)	One-line description	Activity	Input formats	Output formats
deepchecks	https://github.com/deepchecks/deepchecks	~3,800	ML model + data validation suite covering integrity, train/test drift, and model evaluation	Active	Pandas DF + model	HTML report, JSON result objects, <code>CheckResult</code>
Soda Core	https://github.com/sodadata/soda-core	~2,400	YAML-based data quality checks (SodaCL) with 18+ data source connectors	Active (Soda-maintained)	SQL databases, Snowflake, BigQuery, Redshift, Spark, Pandas	YAML/JSON scan result object, Soda Cloud dashboard
TensorFlow Data Validation (TFDV)	https://github.com/tensorflow/data-validation	~800	Schema inference, statistics generation, and anomaly detection for TFX pipelines	Active (Google-maintained)	CSV, TFRecords, Pandas DF	Protocol Buffer (<code>DatasetFeatureStatistics</code>), JSON (via BigQuery integration)
Evidently AI	https://github.com/evidentlyai/evidently	~6,500	Open-source ML/LLM observability with 100+ metrics, reports, and test suites	Active (Evidently-maintained)	Pandas DF, Spark (via API)	HTML report, JSON (<code>report.json</code>), Python dict, Evidently Cloud

PRIMARY — Traditional ML / Statistical Libraries for DQ

Name	URL	Stars (May 2026)	One-line description	Activity	Input formats	Output formats	Strengths vs proje
PyOD	https://github.com/yzhao062/pyod	~8,500+	The most widely used Python library for tabular outlier/anomaly detection (45+ algorithms)	Active (USC team)	Numpy arrays, Pandas DF	Per-sample outlier scores + binary labels (numpy arrays); no structured finding format	Contains ECOD, COPOI IsolationForest, LOF, DBSCAN, One-Class SV autoencoders; PyOD 2 : LLM-guided model sele (ADEngine); benchmarl aligned via ADBench
alibi-detect	https://github.com/SeldonIO/alibi-detect	~2,200	Outlier, adversarial, and concept drift detection with TensorFlow + PyTorch backends	Active (Seldon)	Numpy arrays, Pandas DF, TFRecords	Python DetectorResponse dicts with is_outlier, instance_score, feature_score; JSON-serializable detector configs	Covers KS/Chi-sq/MMD/Wasserstein, drift tests; TabularDr for mixed categorical/numerical columns; detector conf are saveable/loadable
NannyML	https://github.com/NannyML/nannyml	~1,600	Post-deployment model performance estimation and data drift detection without ground truth labels	Active	Pandas DF	Python result objects (.data as DataFrame), interactive Plotly charts	Novel CBPE/DLE algorithm for performance estima without labels; PCA-ba multivariate drift detec links drift alerts back to performance impact
datasketch	https://github.com/ekzhu/datasketch	~2,400	MinHash, LSH, HyperLogLog, and related probabilistic data structures	Active	Python sets, strings (via update())	Python objects (MinHash, MinHashLSH indexes); similarity queries return (key, jaccard_score)	MinHashLSH for sub-li near-duplicate detectic MinHashLSHensem t for containment thresh queries; Redis/Cassan-storage backends for si
scikit-learn outlier modules	https://github.com/scikit-learn/scikit-learn	~60,000+ (full sklearn)	Mature ML library including IsolationForest, LOF, One-Class SVM, EllipticEnvelope	Active	Numpy arrays, Pandas DF	Numpy arrays (decision scores, binary predictions)	Industry standard, well tested, high performan IsolationForest, LOF, DBSCAN implementati directly applicable to L
river	https://github.com/online-ml/river	~5,200	Online ML library for data streams including ADWIN, Page-Hinkley drift detectors	Active	Python iterables / streaming	Python DriftDetector objects (drift flag + statistics)	ADWIN and Page-Hinkl online concept drift; designed for streaming incremental DQ monito

PRIMARY / RARE — LLM-on-top-of-Deterministic-Findings Pattern

⚠ Explicit negative result: As of May 2026, no open-source repository was found that fully implements the four-layer deterministic-engine → structured-findings → LLM-narrative pipeline the project proposes. The following are the closest partial analogues found:

Name	URL	Stars	Description	How close to the pattern	Tag
whylabs/langkit	https://github.com/whylabs/langkit	~600	Text metrics toolkit layered on whylogs for LLM monitoring	Whylogs produces deterministic profiles of LLM I/O text; LangKit adds NLP metrics; WhyLabs platform visualizes. The LLM being monitored is the subject, not the reporter — this is monitoring OF LLMs, not LLM reporting ON data findings	[verified] ^[86] [87] ^[88] [89] [90]

Name	URL	Stars	Description	How close to the pattern	Tag
Acceldata Galileo (closed-source component)	https://github.com/acceldata-ig/(no+public+repo+for+Galileo)	N/A	Acceldata's "Galileo" LLM engine uses GenAI agents (Text-to-Rules, natural-language observability summaries) layered on deterministic profiling and anomaly detection	Closest commercial implementation of the pattern: deterministic DQ engine (profiling + anomaly detection) → structured findings → LLM narrative/recommendations. However, the LLM can also generate SQL and rules autonomously, making it mixed-mode rather than pure interpreter	[verified—commercial] [91][92]
Evidently + custom LLM wrapper	https://github.com/evidentlyai/evidently	~6,500	Evidently produces JSON report snapshots; users can wrap these in an LLM call. No built-in LLM narrative layer.	The JSON output is suitable as Layer-3 input, but no built-in Layer-4 exists in the open-source package. Community examples exist but no maintained repo was found implementing the full pipeline	[verified — gap confirmed] [69][71]

Summary: The deterministic-engine + LLM-narrative architecture as an end-to-end OSS project does not currently exist. The project would occupy a genuine gap.

ADJACENT — One Representative Each

Category	Name	URL	Stars	One-line description	Note
LLM data agent	PandasAI	https://github.com/Sinaptik-AI/pandas-ai	~21,000	Conversational data analysis via LLM code-generation over DataFrames	Representative of what the project explicitly is NOT: LLM explores raw data directly [uncertain — star count]
Auto-visualization	LIDA	https://github.com/microsoft/lida	~4,000	Grammar-agnostic visualization + infographics generation using LLMs	LLM generates and executes visualization code; adjacent, not relevant to DQ finding pipeline [verified] [83][94]
Declarative contracts	frictionless-py	https://github.com/frictionlessdata/frictionless-py	~1,800	DEVT framework for describing, extracting, validating, and transforming tabular data	Frictionless validation report is a candidate baseline for Layer-3 schema design [verified] [95]

Section 3 — Commercial / Closed-Source Tools

Vendor	Tool	One-line description	Overlap with deterministic-engine + LLM-narrative architecture	LLM narrative?	Pricing tier
Monte Carlo	Monte Carlo Data Observability	End-to-end data observability with ML-based anomaly detection, lineage, and RCA	Automated volume/freshness/schema/field-level anomaly detection (deterministic + ML) outputs structured incidents; 2025 "Graph-AI RCA" module generates natural-language root-cause summaries	Yes — incident summaries with NL explanations; closest to Layer-3 + Layer-4	Enterprise SaaS; usage-based
Acceldata	Acceldata Torch + Galileo	Agentic data management platform with AI-driven profiling, DQ, and observability	Galileo LLM engine sits atop deterministic profiling + anomaly detection; "Text-to-Rules" and NL observability summaries; closest to project's architecture conceptually	Yes — NL descriptions, DQ summaries, anomaly explanations, Text-to-Rules; includes guardrails and hallucination QA	Enterprise; contact for pricing
Bigeye	Bigeye	Automated DQ monitoring with adaptive thresholds and SLA management	ML-based adaptive threshold anomaly detection; FlexCheck (YAML/UI rules); 2025 "AI-driven data quality" includes auto-generated check suggestions	Limited — AI assists in check generation; limited narrative reporting	SaaS; usage-based

Vendor	Tool	One-line description	Overlap with deterministic-engine + LLM-narrative architecture	LLM narrative?	Pricing tier
Anomalo	Anomalo	ML-native automated data quality monitoring for warehouses	Unsupervised ML anomaly detection on warehouse tables; root-cause analysis; no deterministic rule definition required	Partial — AI-generated anomaly context summaries [uncertain detail]	Enterprise SaaS
Soda Cloud	Soda Cloud	Cloud extension of Soda Core with AI-assisted check generation and DQ dashboards	SodaAI can suggest SodaCL checks from data structure; cloud dashboards show DQ scores over time	AI-assisted check generation (not post-hoc narrative)	SaaS, free tier for Soda Core OSS
Telmai	Telmai	Real-time AI-powered data quality for data lakes and lakehouses	"Incident Diagnosis Agent" analyzes anomalies and explains root causes; "Data Insight Agent" generates charts and summaries from raw DQ findings; integrates with open table formats	Yes — NL explanations of anomalies and DQ incidents; MCP interface for agents to query validated data context	Enterprise SaaS
Collibra DQ (formerly Owl Analytics)	Collibra Data Quality	Enterprise DQ platform with behavioral rules, anomaly detection, and data catalog integration	Behavioral rules learn from historical data patterns; structured DQ scores per dimension; NL explanation features in enterprise tier	Partial — some NL summaries in 2025 Collibra Intelligence Studio integration [uncertain]	Enterprise
Julius AI / Hex Magic (<i>landscape contrast only</i>)	Julius AI	LLM-first data analysis and visualization agent	LLM explores raw data directly; antithesis of the project's architecture	Yes — but LLM reasons over raw data, not structured findings	SaaS, freemium

Section 4 — Gap Analysis

4.1 Layer-1 and Layer-2 Coverage

Layer-1 (Profiling) — well-covered: The combination of ydata-profiling, DataPrep, and Capital One DataProfiler covers essentially all classical single-column and multi-column profiling metrics defined in Abedjan et al. 2015: type inference, cardinality, null ratios, distributions/quantiles, correlations, duplicate counts, and value-frequency tables. DataProfiler uniquely adds PII entity detection. D-Tale provides interactive exploration but not pipeline-ready JSON output.^{[34][35][36][41][44][45][^]}

Layer-2 (DQ Detection) — partially covered by multiple disjoint tools:

- *Missing-value patterns (MCAR/MAR/MNAR)*: No existing OSS tool systematically classifies missingness mechanisms. ydata-profiling reports null counts; deepchecks flags high null rates; TFDV detects schema violations. Automated MCAR/MAR/MNAR heuristics (e.g., Little's MCAR test, logistic regression-based MAR test) are available in research libraries but not integrated into DQ frameworks.^{[96][97]}
- *Outlier/anomaly detection*: PyOD is the definitive library (45 algorithms, ECOD, COPOD, IsolationForest, LOF). alibi-detect covers drift-based and deep-learning-based outlier detection.^{[76][77][10][9]}
- *Near-duplicate detection*: datasketch for MinHash/LSH; no DQ framework integrates near-duplicate detection natively.^[^84]
- *Distribution/drift testing*: Evidently AI and NannyML cover KS, Wasserstein, PSI, JS divergence. alibi-detect adds MMD and LSDD.^{[79][80][71][72][^76]}
- *Cross-column inconsistency, leakage signals, target imbalance*: deepchecks checks feature-label leakage and train/test distribution; IBM DQ Toolkit adds class overlap and label purity. No single OSS tool covers all of these with a unified finding schema.^{[59][60][^3]}

Summary: Individual algorithms are well-available, but no existing open-source tool assembles all Layer-2 algorithms into a single deterministic engine with a unified finding output format.

4.2 Layer-3 Ontology Gap — Baseline Comparison

The following compares the main candidate schemas for the Layer-3 intermediate representation:

Schema	Source	Structured?	Severity field?	Statistical basis?	LLM-consumable?	Assess
ydata-profiling JSON	<code>profile.to_json()</code>	Yes — per-column statistics dict	No	Yes — distributions, correlations, etc.	Moderate — verbose but parseable	Good Layer baseline; too verbose for LLM consumption without filtering; no issue taxonomy
GE ExpectationValidationResult JSON	<code>to_json_dict()</code>	Yes — {success, expectation_config, result, meta, statistics}	Pass/fail only	Partial — observed vs expected values	Good — clean schema, LLM can parse; but findings are only what was explicitly tested	Strongest schema for finding structure; lacks auto-discovery; severity bias
whylogs profile (JSON summary)	<code>dataset_profile.json</code>	Yes — column-level statistical summaries	No	Yes — counters, histograms, cardinality, frequent items	Moderate — statistics but no issue taxonomy	Best for streaming use; not a full schema, just statistics
Deequ constraint output	Spark DataFrame / JSON conversion	Partial — flat DataFrame rows	Pass/fail	Metric value + threshold	Moderate — needs conversion from Spark DF	Strong for scale; Scala makes PyT integration awkward; flat structure
OpenLineage DQ facets	JSON facets attached to dataset runs	Yes — atomic facets per entity	No	Partial	Good — versioned JSONPointer URIs	Designed for lineage not profiling; it is thin; needs extension for finding on
SodaCL result YAML	Scan result YAML	Yes — per-check pass/fail + metric value	Per-threshold only	Metric value included	Good — YAML is LLM-readable	Clean but declarator dependent; auto-discovery
Frictionless validation report	Python dict / JSON	Yes — per-row/field errors	Error severity (error/warning)	Field name + expected type	Good — clean JSON schema	Schema-declarator focused; not statistical; type/form only

Recommendation: The GE ExpectationValidationResult JSON provides the cleanest existing per-finding schema, but it requires user-defined Expectations rather than auto-discovered issues. The proposed Layer-3 ontology should extend the GE-style schema with: (a) an `issue_type` enum (drawn from DAMA-DMBOK 6 dimensions), (b) an auto-discovery provenance field (`detected_by`: algorithm name), (c) a `severity` field grounded in the statistical basis, and (d) an `ml_impact` array. The OpenLineage DQ facet pattern provides a useful versioned-URI model for `_schemaURL` fields.^{[46][98][47][99]}

4.3 Layer-4 Gap — LLM as Interpreter

Where this pattern exists (partially):

- Acceldata Galileo and Telma implement commercial versions where an LLM interprets structured DQ findings and generates NL summaries. However, both allow the LLM to also generate SQL and rules autonomously (mixed-mode, not pure interpreter).^{[9][92][100][101]}
- PyOD 2's LLM-guided model selection is a narrow instance of the pattern: LLM reads algorithm metadata + task description → selects a model. No raw data is exposed.^{[10][11]}
- The XAI narrative literature (LLMs explaining SHAP values, counterfactuals) demonstrates the technical feasibility of LLM interpretation of structured ML outputs, but is not applied to tabular DQ specifically.^{[102][25]}

Where it is absent:

- No open-source tool implements a full "DQ engine → Layer-3 findings → LLM narrative report" pipeline.
- The LangKit/whylogs stack monitors LLMs, it does not use LLMs to narrate tabular DQ findings.^{[86][88]}
- Evidently's JSON output is structurally compatible with Layer-3 consumption, but no built-in LLM layer exists.^{[71][72]}

Key risk confirmed by literature: Even with structured input, LLMs exhibit hallucination on numerical reasoning. The faithfulness review confirms that RAG-style strict grounding (providing only the Layer-3 JSON as context)

combined with structured output formats significantly reduces but does not eliminate hallucination. Evaluation using faithfulness metrics (FactScore, NLI-based entailment) against Layer-3 findings should be part of the project's Layer-4 evaluation strategy.^{[103][26]}

4.4 Combined-Pipeline Gap

Tool	Layer 1	Layer 2	Layer 3	Layer 4
ydata-profiling	✔ Full	✗	Partial (JSON stats)	✗
Great Expectations	Partial (profiler)	Partial (expectations)	✔ (per-finding JSON)	✗
Evidently AI	✔ (Data Quality preset)	✔ (drift, missing, outlier stats)	✔ (JSON report)	✗ (no LLM layer)
deepchecks	Partial	✔ (ML-context checks)	Partial (CheckResult)	✗
PyOD	✗	✔ (outlier detection)	✗	✗
Acceldata (commercial)	✔	✔	✔	✔ (partially — mixed-mode LLM)
Monte Carlo (commercial)	✔	✔	✔	Partial (RCA summaries)
Project (proposed)	✔	✔	✔ (novel ontology)	✔ (strict interpreter-only)

No single open-source tool covers all four layers. Evidently AI is closest for Layers 1–3; the project's unique value is the complete pipeline with a strict, grounded Layer-4.

4.5 DQ-Dimensions Grounding Gap

Most tools grade severity against **ad-hoc thresholds** (e.g., null rate > 0.2 = warning), not against recognised frameworks. The IBM DQ Toolkit is the only reviewed tool that explicitly grounds findings in ML-specific quality dimensions. The BIR 2025 review confirms: across 151 tools reviewed, only 10 support AI-augmented DQM, and explicit DAMA-DMBOK or ISO 25012 grounding in severity scores is rare. The project's explicit commitment to standard-grounded severity (DAMA-DMBOK 6 dimensions, ISO/IEC 25012 fitness-for-purpose) as the severity calibration layer is a differentiator.^{[20][21][104][17][3][32]}

4.6 Strongest Differentiator

The combination of three properties is not found in any existing OSS tool:

- 1. **Auto-discovery + auto-severity** in Layer 2 — detecting DQ issues without requiring user-declared rules or Expectations
- 2. **Structured finding ontology** in Layer 3 — a typed, severity-graded, ML-impact-annotated JSON schema that can serve as an LLM trust anchor
- 3. **LLM-as-interpreter-only** in Layer 4 — the LLM narrates Layer-3 findings but cannot compute statistics, generate code, or access raw data

This combination allows the report to be **reproducible** (same Layer-3 input → stable narrative) and **auditable** (every LLM claim cites a Layer-3 finding). No commercial tool fully achieves this; Acceldata Galileo comes closest but retains LLM code-generation autonomy.

4.7 Known Technical Risks Not Solved by Existing Tools

- 1. **LLM hallucination on numerical findings** even with **structured input**: Faithfulness literature confirms LLMs still fabricate numbers when summarising tables. Mitigation: strict templated prompts with exact value injection, output validation against Layer-3 schema, and faithfulness-score gating before report delivery.^{[26][103]}
- 2. **Severity calibration when multiple issues compound**: No tool handles compound severity (e.g., high missingness + high cardinality + distribution drift in the same column together indicate higher risk than each alone). Current tools score findings independently. The project's Layer-3 ontology should include a `compound_severity` field and logic for escalation.
- 3. **Scale of profiling on wide tables (>1,000 columns)**: ydata-profiling becomes slow beyond ~200 columns; Deequ on Spark handles this better but requires Spark infrastructure. DataPrep's Dask backend is the best OSS solution at scale for Python-native users.^{[41][51][^53]}
- 4. **Semi-structured columns (JSON-in-text, free-text fields)**: No reviewed tool handles embedded JSON in text columns as part of a profiling pipeline. Capital One DataProfiler handles multiple file formats but not intra-column JSON. This remains an unsolved problem in the OSS landscape.^{[44][45]}

5. Mapping a finding to a Data/AI-readiness verdict: The IBM DQ Toolkit makes the closest attempt, but its metrics are limited. The project needs an explicit "trainability verdict" layer (can this dataset train an unbiased ML model?) that no current tool provides end-to-end.^[^3]
6. Reproducibility of the LLM narrative across runs: LLMs with non-zero temperature produce different narratives on identical Layer-3 inputs. Mitigation: temperature=0 for report generation, deterministic seed where available, and hash of Layer-3 findings stored alongside report for audit.

Section 5 — Suggested Reading Order

1. Abedjan, Golab, Naumann (2015) — "Profiling Relational Data: A Survey" (VLDB Journal, DOI 10.1007/s00778-015-0389-y): Read first. This paper defines the complete taxonomy of profiling tasks that Layer 1 must implement. Every statistic computed in the project maps to a concept defined here. Without this foundation, Layer-1 design will reinvent imprecisely.^{[2][1]}
2. Zhou et al. (2024) — "A Survey on Data Quality Dimensions and Tools for Machine Learning" (arXiv:2406.19614): Read second. This is the most direct survey of what DQ tools currently cover and where the ML-contextual gap lies. The four-dimension taxonomy and the coverage analysis of 17 tools directly informs which Layer-2 checks to prioritize and what Layer-3 severity fields mean.^{[17][18][^16]}
3. Dong, Sahri, Palpanas (2024) — "Data Quality Awareness: A Journey from Traditional Data Management to Data Science Systems" (arXiv:2411.03007): Read third. Bridges classical DQ (where DAMA dimensions and ISO standards originate) to modern ML DQ challenges. Provides the intellectual lineage needed to ground Layer-3 severity in recognised standards rather than ad-hoc thresholds.^{[15][14]}
4. Gupta et al. (2021) — "Data Quality Toolkit: Automatic Assessment of Data Quality and Remediation for Machine Learning Datasets" (arXiv:2108.05935): Read fourth. The most direct paper-level prior art for Layers 2 + 3. Introduces ML-specific DQ metrics (class overlap, feature redundancy, label noise) not covered by classical profiling surveys, and provides an API-accessible result structure to learn from and extend.^[^3]
5. Han et al. (2022) — "ADBench: Anomaly Detection Benchmark" (NeurIPS 2022, arXiv:2206.09426): Read fifth. The benchmark that empirically validates algorithm choices for Layer 2's outlier detection engine. Required reading before committing to a specific algorithm configuration (IsolationForest vs. ECODE vs. LOF vs. COPOD) for different data regimes.^{[5][6][^4]}

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4. [ADBench: Anomaly Detection Benchmark - OpenReview](#) - This paper presents ADBench, a comprehensive tabular anomaly detection benchmark with 30 algorithms ...
5. [Benchmarks - pyod 3.2.1 documentation](#) - ADBench [AHHH+22] is a 45-page study evaluating 30 anomaly detection algorithms on 57 tabular benchm...
6. [ADBench: Anomaly Detection Benchmark - NeurIPS](#)
7. [ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions](#) - Outlier detection refers to the identification of data points that deviate from a general data distr...
8. [4 Empirical Evaluation](#) - Outlier detection refers to the identification of data points that deviate from a general data distr...
9. [\[2412.12154\] PyOD 2: A Python Library for Outlier Detection ... - arXiv](#) - In this paper, we demonstrate how PyOD 2 streamlines the deployment and automation of OD models and ...
10. [PyOD 2: A Python Library for Outlier Detection with LLM ...](#)
11. [PyOD 2: A Python Library for Outlier Detection with LLM-powered Model Selection](#)
12. [Data-centric Artificial Intelligence: A Survey](#)
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15. [\[2411.03007\] Data Quality Awareness: A Journey from Traditional ...](#) - In this paper, we present a comprehensive review of the evolution of data quality (DQ) awareness fro...
16. [\[PDF\] A Survey on Data Quality Dimensions and Tools for Machine Learning](#) - In this survey, we review 17 DQ evaluation and improvement tools in the last. 5 years. By introduc...

17. [\[Literature Review\] A Survey on Data Quality Dimensions and Tools ...](#) - The paper discusses the correlation between data quality and machine learning efficacy. It elaborate...
18. [A Survey on Data Quality Dimensions and Tools for Machine Learning](#) - The study revealed that most tools focus on intrinsic and representational data quality, with less a...
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